RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
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DEPARTMENT OF MATHEMATICS

UNIT – 1 - VECTOR DIFFERENTIATION

PART – A

1. Find unit normal vector of the surface $z = x^2 + y^2$ at (-1, -2, 5)
2. Compute the Laplacian of the function $\phi = x^2 + y^2 + z^2$
3. Compute the Curl (xy i + yz j + zx k).
4. A unit tangent vector to the surface $x = t$, $y = t^2$, $z = t^3$ at $t = 1$
5. If $u = x^2 y^2$ and $v = xy - 3z^2$, then determine the value of $\nabla.(\nabla u \times \nabla v)$.
6. If $\vec{r} = t^3\hat{\imath} + (2t^3 - \frac{1}{5t^2})\hat{\jmath}$, show that $(\vec{r} \times \frac{d\vec{r}}{dt})$ is a constant vector.

PART – B

1. A fluid motion is given by $\vec{v} = (y + z)\hat{\imath} + (z + x)\hat{\jmath} + (x + y)\hat{k}$. Show that the motion is irrotational and hence find the velocity potential.
2. A particle moves along the path $\vec{r} = \cos(\omega t)\,\hat{\imath} + \sin(\omega t)\hat{\jmath}$, where '$\omega$' is a constant. Determine the velocity, acceleration vectors at any instant of time 't'. Further show that $\vec{v}$ is perpendicular to $\vec{r}$ and $(\vec{r} \times \vec{v})$ is a constant vector.
3. A fluid motion is given by $\vec{v} = (y\,sinz - sinx)\hat{\imath} + (x\,sinz + 2yz)\hat{\jmath} + (xy\,cosz + y^2)\hat{k}$. Is the motion irrotational? If so, find the velocity potential.
4. Show that the vector field $\vec{F} = \frac{\vec{r}}{|r|^3}$ is conservative as well as solenoidal. Find the scalar potential.
5. Show that $iv(grad\, r^n) = n(n+1)r^{n-2}$, where $r = \sqrt{x^2 + y^2 + z^2}$. Hence show that $\nabla^2\left(\frac{1}{r}\right) = 0$.
6. Find the value of n for which the vector $r^n\vec{r}$ is solenoidal, where $\vec{r} = x\hat{\imath} + y\hat{\jmath} + z\hat{k}$.
7. If $\vec{F} = (x + y + 1)\hat{\imath} + \hat{\jmath} - (x + y)\hat{k}$, show that $\vec{F}$ is orthogonal to $curl(\vec{F})$.
8. Obtain the directional derivative of $\psi = xy^2 + yz^3$ at $(2, -1, 1)$ in the direction of normal to surface $xlog(z) - y^2 = -4$ at (-1, 2, 1).
9. Find the values of 'a' and 'b' so that the surfaces $ax^2 - byz = (a + 2)x$ and $4x^2y + z^3 = 4$ may cut orthogonally at the point $(1, -1, 2)$.
10. If $\overrightarrow{F} = (3x^2y - z)\,i + (xz^3 + y)\,j - 2x^3z^2k$, then find div ($\vec{F}$) and curl ($\vec{F}$) when (1, -1, 2). Is $\overrightarrow{F}$ solenoidal or irrotational? Justify your answer.
11. Find the divergence of the field $\frac{xi + yj + zk}{(x^2 + y^2 + z^2)^{3/2}}$. Is this field solenoidal?
12. Find the constants m and n such that the surface $2x^2 - 2y^2 - 3z^2 + 8 = 0$ is orthogonal to $mx^2 + y^2 = nz$ at the point (-1, 2, 1).

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UNIT – 2 - VECTOR INTEGRATION

PART – A

1. The value of the line integral $\int \text{grad}(x+y-z)\overrightarrow{dr}$ from (0, 1, -1) to (1, 2, 0) is _____.
2. The value of the surface integral $\iint_S (yzdydz + zxdzdx + xydxdy)$ where S is the surface of the sphere $x^2 + y^2 + z^2 = 1$.
3. The work done by a force $y\hat{\imath} + x\hat{\jmath}$, which displaces a particle from the origin to point $\hat{\imath} + \hat{\jmath}$ is _____.
4. If $\vec{F}$ is the velocity of a fluid particle then $\int_C \vec{F}.d\vec{r}$ represents _____.
5. Let S be a closed orientable surface enclosing a unit volume. Then the magnitude of the surface integral $\iint_S \vec{r}.\hat{n}ds$, where $\vec{r} = x\hat{\imath} + y\hat{\jmath} + z\hat{k}$ and $\hat{n}$ is the unit normal to the surface S is _____.

PART – B

1. Find the work done in moving a particle in the force field $\vec{F} = (3x^2)\hat{\imath} + (2xz - y)\hat{\jmath} + z\hat{k}$, along the curve $x^2 = 4y$ and $3x^3 = 8z$ from $x = 0$ to $x = 2$.
2. Evaluate $\int_C \vec{F}.d\vec{r}$ where $\vec{F} = (x^2)\hat{\imath} + (xy)\hat{\jmath}$ and C is the boundary of the square in the plane $z = 0$ and bounded by the lines $x = 0$, $y = 0$, $x = a$ and $y = a$.
3. Evaluate $\int_S \vec{F}.\hat{n}dS$, where S is the part of the surface of the sphere $x^2 + y^2 + z^2 = a^2$ in the first octant, and $\overrightarrow{F} = yz\hat{\imath} + zx\hat{\jmath} + xy\hat{k}$.
4. Evaluate $\int_S \vec{F}.\hat{n}dS$, where S is the part of the surface of the plane $2x + y + 2z = 6$ which lies in the positive octant, and $\vec{F} = 4x\hat{\imath} + y\hat{\jmath} + z\hat{k}$.
5. Verify Green's theorem for $\int_C (xy + y^2)dx + x^2dy$, where C is the path bounded by the line $y = x$ and the parabola $y = x^2$.
6. Evaluate $\int_C (3x^2 - 8y^2)dx + (4y - 6xy)dy$, where C is the boundary of the region enclosed by the lines $x = 0, y = 0, x + y = 1$
7. Verify Stokes' theorem for the line integral of $\vec{f} = (x^2 + y^2)\hat{\imath} - 2xy\hat{\jmath}$, taken around the rectangle bounded by the lines $x = \pm a, y = 0, y = b$.
8. Evaluate $\int_C \vec{F}.d\vec{r}$ using Stokes' theorem where $\overrightarrow{F} = y\hat{\imath} + z\hat{\jmath} + x\hat{k}$, where C is the rim for the upper part of the sphere $x^2 + y^2 + z^2 = a^2$
9. Verify divergence theorem for $\overrightarrow{F} = (2x - z)\hat{\imath} + x^2y\hat{\jmath} - xz^2\hat{k}$ for the region bounded by the planes $x = 0, x = 1, y = 0, y = 1, z = 0, z = 1$.
10. Evaluate $\int_S \vec{F}.\hat{n}dS$, where S is the part of the surface of the cylindrical region bounded by $x^2 + y^2 = 9, z = 0$ and $z = 2$, $\overrightarrow{F} = y\hat{\imath} + x\hat{\jmath} + z^2\hat{k}$.

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UNIT – 3 - PARTIAL DIFFERENTIAL EQUATIONS

PART – A

1. Find the partial differential equation by eliminating a and b from the relation u = $ax + by + ab$.
2. Find the partial differential equation whose solution is f (3z, xy) = 0.
3. Derive a partial differential equation of all spheres of fixed radius having centers in the yz-plane.
4. The partial differential equation formed by eliminating the arbitrary constants $c, d$from the relation $z = (x + c)(y + d)$ is_____.
5. The partial differential equation formed by eliminating the arbitrary constants m, n from the relation $mz + n = m^2x + y$ is_____.
6. The partial differential equation formed by eliminating the arbitrary function g from the relation $z = e^y g(x + y)$ is_____.

PART – B

1. Form the partial differential equation by eliminating the arbitrary constants from $\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1$.
2. Form the partial differential equation by eliminating the arbitrary functions in the relation $z = f(y + 2x) + g(y - 3x)$.
3. Form the partial differential equations:
   i. The partial differential equation formed by eliminating the arbitrary constants m, n from the relation $mz + n = m^2x + y$.
   ii. The partial differential equation formed by eliminating the arbitrary constants a, b from the relation $z = (x^2 + a)(y^2 + b)$.
   iii. The partial differential equation formed by eliminating the arbitrary constants a, b from the relation $2z = \frac{x^2}{a^2} + \frac{y^2}{b^2}$.
   iv. The partial differential equation formed by eliminating the arbitrary function g from the relation $z = e^y g(x + y)$.
4. Solve : $\left(\frac{y-z}{yz}\right)p+\left(\frac{z-x}{zx}\right)q=\left(\frac{x-y}{xy}\right)$.
5. Solve : $\left(x^2-yz\right)p+\left(y^2-zx\right)q=\left(z^2-xy\right)$
6. Obtain the various possible solutions of one-dimensional wave equation $u_{tt} = c^2 u_{xx}$ by the method of separation of variables.

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7. Discuss the various possible solutions of the diffusion equation $\frac{\partial z}{\partial t} = c^2 \frac{\partial^2 z}{\partial x^2}$ using the separation of variables method.

8. Form a partial differential equation by eliminating arbitrary constants 'a' and 'b' from $(x - a)^2 + (y - b)^2 + z^2 = 4$.
9. Discuss the various possible solutions of two-dimensional Laplace equation.
10. Obtain the various possible solutions of one-dimensional heat equation $u_t = c^2 u_{xx}$ by the method of separation of variables.
11. By the method of separation of variables, solve the equation $\frac{\partial z}{\partial x} = 2 \frac{\partial z}{\partial t} + u$, $u(x, 0) = 6 e^{-3x}$

12. By the method of separation of variables, solve the equation $x^2 \frac{\partial^2 z}{\partial x \partial y} + 3 y^2 z = 0$.

13. By the method of separation of variables, solve the equation $\frac{\partial^2 z}{\partial x^2} - 2\frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} = 0$.

UNIT – 4 - NUMERICAL METHODS - I

PART A

01. The root of the equation $x \log_{10} x = 1.2$ lies in the finite interval _____.

02. Perform two iterations of Newton-Raphson method to obtain the value of $\sqrt{5}$ with initial approximation as 2.
03. First approximation to the root of the equation $x \tan x + 1 = 0$ by Newton-Raphson method near $x = 2.5$ is_____.

04. The first approximation of a real root of the equation $xe^x = 3$ lying in $(1, 2)$ using method of false position is _____.
05. The approximate solution of $\frac{dy}{dx} = x^2$, $y(1) = 1$, by Taylor's series method upto second degree term is _____.
06. The Taylor series solution of $y' = x^2 + y^2$, $y(0) = 1$ up to second degree terms is_____.

PART B

01. Using the method of false position find an approximate real root of $3x - \cos x = 4$ correct to four places of decimals.

02. Find the real root of the equation $x^4 - x - 10 = 0$ by using regula falsi method between 1 and 2. Consider upto four decimal places.

03. Using method of false position find a positive real root of the equation $x\sin(x) + \cos(x) = 0$ which lies between 2 and 3 correct three places of decimal.

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04. The biologist have arrived at the formula to find bacteria concentration C in reservoirs at different time t given by $C = 4e^{-2t} + e^{-0.1t}$. Find the actual time very close to $t = 7$ such that the bacteria concentration is 0.5 by applying Newton-Raphson method.

05. Newton-Raphson method is an iterative process which gives better approximation towards the root of an equation provided a value close to the solution is known. Use this method to find a solution of the transcendental equation $e^x = x^3 + \cos(25x)$ near 4.5, correct to four decimal places.

06. Apply Taylor's series method to obtain $y(0.1)$ considering up to fourth degree term, if $y(x)$ satisfies the equation $\frac{dy}{dx} = x - y^2$, $y(0) = 1$.

07. Expand $f(x) = e^{2x}$ in ascending powers of $(x-1)$ up to term containing $(x-1)^4$, using Taylor's series method hence compute $e^{1.1}$.

08. Given initial value problem $\frac{dy}{dx} = 3e^x + 2y$, $y(0) = 0$. Compute $y(0.1)$ by taking $h = 0.1$ using the Runge-Kutta method of fourth order.

09. Using fourth order Runge-Kutta method find y at $x = 0.2$ taking step size as 0.1. Given that , $\frac{dy}{dx} = 3e^x + 2y$, $y(0) = 0$.

10. Using fourth order Runge-Kutta method, compute y at $x = 1.1$ and $x = 1.2$ from $\frac{dy}{dx} = \frac{(1+x)}{(1+y)}$, $y(1) = 2$, $h = 0.1$. Correct to 3 decimal places.

11. Apply Milne's method to solve the differential equation $y' = x - y^2$, for $x = 0.8$ Given that:

x	0	0.2	0.4	0.6
y	0	0.02	0.0795	0.1762

Apply the corrector twice, (with $h = 0.2$).

12. Using Milne's predictor-corrector method to compute $y(1.4)$, given that $\frac{dy}{dx} = x^2(1+y)$, $y(1) = 1$, $y(1.1) = 1.233$, $y(1.2) = 1.548$, $y(1.3) = 1.979$.

UNIT – 5 - NUMERICAL METHODS - II

PART A

01. Construct the forward difference table for the following data points:

x_i	2	4	6	8
y_i	3	8	12	7

02. Construct the backward difference table of the polynomial $y = x^3 - 3x + 1$ for $x = -1, 0, 1$.

03. Evaluate: $\Delta^3((2x^2+1)(x+1))$, by taking $h = 2$

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04. If $y(0) = 0, y(1) = 2, y(2) = 2.5, y(3) = 2.3, y(4) = 2, y(5) = 1.7, y(6) = 1.5$, then, by Weddle's rule, $\int_0^6 y\,dx$ =_____.
05. If $y(0) = 0, y(1) = 2, y(2) = 2.5, y(3) = 2.3, y(4) = 2, y(5) = 1.7, y(6) = 1.5$, then, by Simpson's three-eighth rule, $\int_0^6 y\,dx$=_____.
06. Compute $y'(-1)$ for the following data points:

x_i	-1	0	1
y_i	3	6	12

PART B

01. The following data defines the sea-level concentration of dissolved oxygen for fresh water as a function of temperature:

T^oC	0	8	16	24	32
O, mg/L	14.621	11.843	9.870	8.418	7.305

Using Newton-Gregory formula, calculate

(i) the amount of oxygen when temperature 10^oC and35^oC.
(ii) the rate of change of the amount of oxygen at temperature $T = 8$℃.

02. Bessel functions often arise in advanced engineering and scientific analyses such as the study of electric fields. These functions are usually not amenable to straightforward evaluation and therefore are often compiled in standard mathematical tables. One such data is given below:

x	0	0.5	1.0	1.5	2.0
$J_1(x)$	0	0.2422	0.4400	0.5579	0.5767

If $J_1(x)$ represents Bessel function, calculate

(i) $J_1(0.8), J_1(2.1)$ using Newton-Gregory interpolation formulae.
(ii) $\frac{dJ_1}{dx}$ and $\frac{d^2J_1}{dx^2}$ at $x = 1.0$.

03. Details regarding marks secured by 280 candidates in an examination are given by the following table. Using Newton's forward difference interpolation formula estimate the number of candidates who secured marks between 45 and 50.

Marks	Below 30	30-40	40-50	50-60	60-70	70-80
No. of candidates	35	49	62	74	40	20

04. The following table gives the values of observed temperature θ(in degree centigrade) of a vessel of cooling water at different times t(in minutes).

t	1	3	5	7	9
θ	85.3	74.5	67.0	60.5	54.3

Estimate (i) the temperature at 7.5. (ii) the approximate rate of cooling at t = 1 & t = 7.

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05. The population of a certain city is taken from census records and is given below

Year	1980	1990	2000	2010	2020
Population (in lakhs)	40.62	60.80	71.95	103.56	132.65

Estimate (i) The projected population in 2025.

(ii) The rate of growth of population in 2010.

06. The following table gives the weights of babies during nine months of life.

Age (in months)	0	3	5	9
Weights (in pounds)	5	12	15	22

Obtain the cubic polynomial to represent the weights of the babies and estimate weight at the age of seven months using Lagrange's interpolation formula.

07. Given the data $(1, 2), (3, 5), (7, 12), (13, 6)$ using suitable interpolation method find the cubic polynomial.

08. Distances in nautical miles of the visible horizon for given heights in metres given above the surface of the earth are given by the following table:

x(height)	100	150	250	300
y(distance)	12	18	27	36

Determine the value of distance at the height 125 metres and 252 metres using Lagrange's interpolation.

09. Evaluate $\int_0^{\frac{\pi}{2}} \sqrt{sinx + cosx}\, dx$, by applying Simpson's 1/3 rule, Simpson's 3/8 rule and Weddle's rule.

10. Find an approximate value of $\int_0^3 \ln(1 + x^2)\, dx$, by considering seven ordinates of the interval (0, 3) using Simpson's 1/3 rule, Simpson's 3/8 rule and Weddle's rule.

11. Find an approximate value of $\int_0^1 e^{-x^2}\, dx$, by considering seven ordinates of the interval (0, 1) using Simpson's 1/3 rule, Simpson's 3/8 rule and Weddle's rule.

12. The table below gives the areas of cross sections mentioned in a road project earth work computation at intervals of 10 metres.

$x(m)$	0	10	20	30	40	50	60
$A(m^2)$	126	248	118	90	320	120	75

Calculate the volume of the earth work required using Simpson's 1/3 rule, Simpson's 3/8 rule and Weddle's rule.